Auction-Based Online Policy Adaptation for Evolving Objectives

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Abstract

We consider multi-objective reinforcement learning problems where objectives come from an identical family—such as the class of reachability objectives—and may appear or disappear at runtime. Our goal is to design adaptive policies that can efficiently adjust their behaviors as the set of active objectives changes. To solve this problem, we propose a modular framework where each objective is supported by a selfish local policy, and coordination is achieved through a novel *auction*-based mechanism: policies bid for the right to execute their actions, with bids reflecting the urgency of the current state. The highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives. Going back to the original adaptation problem, when objectives change, the system adapts by simply adding or removing the corresponding policies. Moreover, as objectives arise from the same family, identical copies of a parameterized policy can be deployed, facilitating immediate adaptation at runtime. We show how the selfish local policies can be computed by turning the problem into a general-sum game, where the policies compete against each other to fulfill their own objectives. To succeed, each policy must not only optimize its own objective, but also reason about the presence of other goals and learn to produce calibrated bids that reflect relative priority. In our implementation, the policies are trained concurrently using proximal policy optimization (PPO). We evaluate on Atari Assault and a gridworld-based path-planning task with dynamic targets. Our method achieves substantially better performance than monolithic policies trained with PPO.

1 Introduction

Consider the problem of controlling a mobile robot deployed in a cat shelter whose task is to deliver food to cats roaming around the facility (Figure 1). New food requests from hungry cats may arrive at arbitrary times and locations, while existing requests may disappear if a cat wanders away before being served. Since the robot can serve only one cat at a time, fulfilling one request may delay others, potentially allowing some cats to move farther away. Thus, the robot must dynamically adapt to the set of active requests to maximize the number of cats fed before they disappear.

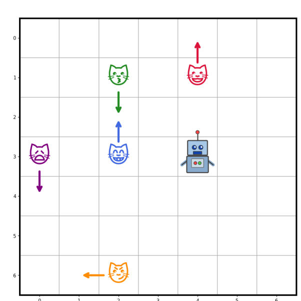


Figure 1: Cat Feeder env.

In this work, we model the above control problem as a multi-objective reinforcement learning task in which all the objectives come from an identical family, such as reachability in this case. These objectives can be conflicting and any number of them may be active at a given time. Moreover, we make no assumptions about their distribution or arrival pattern. Our goal is to design adaptive policies that update their behavior seamlessly as objectives appear or disappear. This is a natural problem relevant to many real-world scenarios and has been extensively studied in the planning literature [9], typically assuming that a precise environment model (like a graph) is available and there is no uncertainties. In contrast, within reinforcement learning (RL) settings—where the environment may be uncertain and a model may not be available—this problem has not been studied to the best of our knowledge.

To solve this problem, we propose a compositional RL framework, where each objective is served using a local policy, selfishly maximizing its assigned objective by competing against the other policies. Coordination between these policies is achieved through a novel auction-based mechanism: at every decision step, policies submit bids for the right to execute their actions, where bids reflect the urgency of the current state with respect to their objectives. The highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives. Returning to the cat feeder example, suppose the robot approaches an intersection with two active requests, the policy for the first request recommends turning left, while the policy for the second recommends turning right. Since both actions cannot be executed simultaneously, a trade-off is unavoidable. The auction mechanism determines which objective is prioritized, based on the relative urgency encoded in the bids.

This compositional design yields strong modularity, facilitating fast adaptation: when an objective appears or disappears, only the corresponding policy needs to be added or removed. Furthermore, when objectives belong to the identical family—which is reachability in our case—a single generalized policy can be trained for the entire family. We can deploy identical *copies* of this policy when new objectives arrive, enabling immediate adaptation. Modularity also permits us to define objective-specific reward signals while the auction mechanism takes care of the trade-offs between objectives. For example, we can easily define a distance-based dense reward signal for each policy controlling the cat feeder to move closer to its assigned cat. In contrast, it is challenging to define such a dense reward signal to train a monolithic policy to trade-off between the multiple objectives. We have to resort to imposing a preference ordering over the multi-dimensional reward vector, or stick to sparse rewards. We show in our experiments that monolithic policies, trained with both sparse rewards and different types of scalarized dense rewards, fall much short of the payoffs achieved by our modular approach.

Yet another advantage of our modular framework is interpretability. At any moment, we can determine the policy in control, and thereby the objective currently being pursued by the agent. This explanation is beneficial in a range of real-world situations, like providing post-hoc explanations of failures, and also feedback on biases in pursuing different objectives.

We implement our modular framework as a general-sum game between agents that control the local policies to fulfill their assigned objectives, and this game is set up using a simple wrapper around existing multi-objective environments. Each agent's action space is augmented with an additional numeric variable representing the bids, and the transitions between the game's states encode the execution of the action selected by the highest bidder. However, we still need to address some challenges.

Challenge I: Enforcing honest bids. While the auction mechanism treats all policies fairly, the policies must still place honest bid in proportion to their urgency. Otherwise, they may learn to overbid and suppress competitors, resulting in suboptimal results. To prevent this, we impose penalties during training that scale with the bid values. As a consequence, excessive bidding reduces the net return of a policy, thus aligning its incentives with truthful urgency estimation—a policy only bids high when it is sure to receive a reward (when it feeds the cat) or to avoid a penalty (when a cat is close to disappearing).

Challenge II: Achieving environment awareness. Effective bidding depends not only on the current state and the local objective, but also on the configuration of competing objectives. For instance, in the cat feeder problem, if two requests are spatially close, their policies need not compete aggressively. Conversely, if requests lie in opposite directions, the urgent policy must bid sufficiently high to win. To enable such environment-aware bidding, we allow the policies to observe all the objectives, i.e., the policies have information about the global state. Additionally, the policies are trained concurrently so that they learn to optimize payoffs in the presence of opponents.

Challenge III: Unbounded objective count. When policies have to compete with many objectives, providing each with the global state results in a very high dimensional observation space. This makes training slow with a standard neural network policy. We address this with an attention pooling mechanism that aggregates information about an arbitrary number of competing objectives into a fixed-dimensional representation. Each local policy is equipped with its own pooling module which is co-trained with the policy. A bonus of this architecture is that it allows policies to compete with more objectives than those during training, since the pooling mechanism can extract useful features from an arbitrary number of objective states.

We test our framework on two multi-objective environments: the cat feeder environment implemented as a gridworld, and the Atari Assault game from the Arcade Learning Environment [6]. To train our policies, we take inspiration from **(author?)** [27] who provide guidance on using proximal policy optimization (PPO; **(author?)** [23]) in multi-agent systems. We observe empirical convergence of training and substantially better performance compared to baselines in both environments. Lastly, to aid interpretability in renderings of rollouts, we implement visualization of the policy in control and the objective being pursued.

Related Work

A large body of work in multi-objective reinforcement learning (MORL) relies on *scalarization*, where multiple reward functions are aggregated into a single scalar objective so that standard RL algorithms can be applied. The simplest approach uses weighted sums of rewards [10], though nonlinear scalarization methods have also been proposed [25]. A key limitation is that the induced importance among rewards may not reflect the designer's intent, often leading to a tedious debugging process [11]. In contrast, our approach achieves a trade-off among reward components without collapsing them into a fixed scalar objective.

Other works address trade-offs by adopting a specific optimality notion, such as Pareto optimality [26] and its approximations [20], or fairness-based criteria across rewards [19, 7, 24]. These approaches typically learn a single monolithic policy satisfying the chosen criterion. By contrast, we learn independent, selfish policies for each reward component and compose them at runtime, preserving modularity while achieving a coherent global trade-off.

Only a few works consider distributed policies for multiple rewards. W-learning [14] and its deep RL extension [21] train separate policies alongside meta-functions that score states by urgency, selecting the policy with the highest score at runtime. Other methods aggregate decisions through ranked voting [17] or fixed rules such as summing action values [22]. Our approach is simpler: it uses an engineered reward structure that enables standard RL algorithms (e.g., PPO) without additional meta-policies or complex aggregation. Moreover, to the best of our knowledge, we are the first to study the incremental MORL setting, where reward components can be added or removed at runtime.

The idea of bidding-based selfish policies is inspired by auction mechanisms for multi-objective path planning on graphs [4] and by the literature on bidding games [15, 2, 3]. These works use explicit model-based approaches and do not readily extend to the model-free setting of RL. Besides, they are restricted to the setting with two objectives.

2 Multi-Objective Discounted Reachability with Deadlines

We formalize the policy synthesis problem for a fixed number of goals, and then introduce our auction-based framework for this setting. Afterwards, we will describe how our framework naturally extends to the setting with dynamic goal appearance and disappearance.

Policy synthesis for a fixed goal set. We use multi-objective Markov decision processes (MO-MDP) to model uncertain environments with a *fixed* number of discounted reachability objectives with deadlines. An MO-MDP with $m \in \mathbb{N}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, D, R, \gamma, M, \mu)$ where S is the set of *locations* equipped with the Euclidean metric $d(\cdot, \cdot)$, with $\mathbf{S} = S^{m+1}$ being the *state space* such that $(s_0, s_1, \ldots, s_m) \in \mathbf{S}$ includes the agent's location s_0 and the location of the i-th goal for every i, A is the set of *actions*, $T : \mathbf{S} \times A \to \Delta(\mathbf{S})$ is the probabilistic *transition function*, $D = \{d_i \in \mathbb{N} \cup \infty\}_{i \in [1;m]}$ is the set of goal-specific *deadlines* ($d_i = \infty$ effectively means the i-th goal has no deadline), and without loss of generality, we assume $d_1 \leq \ldots \leq d_m$, $R = \{r_i \in \mathbb{R}_{>0}\}_{i \in [1;m]}$ is the set of *rewards* obtained after the completion of each goal, $\gamma \in (0, 1)$ is the *discount factor*, $M > r_{\max} := \max_{i \in [1;m]} r_i$ is the per-objective one-time *penalty* for missing the deadline,**KM:** the penalty is always added undiscounted and $\mu \in \mathbb{D}(\mathbf{S})$ is the *initial state distribution*.

A *policy* in an MO-MDP $\mathcal{M}$ is a function $\pi : (\mathbf{S} \times A)^* \times \mathbf{S} \to \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. **KM:** Restrict to memoryless policies? The policy π is executed until the last deadline d_m, giving rise to a random sequence of states and actions $\rho = s^0, a^0, s^1, a^1, \ldots, s^{d_m} a^{d_m}$, called a *trace*. For each $i \in [1; m]$, let τ_i be the first time point at which the i-th goal is reached, i.e., $s_0^{\tau_i} = s_i^{\tau_i}$, and $\tau_i = \infty$ if the i-th goal is never reached. The per-objective reward collected on the trace ρ is $w_i(\rho) = \gamma^{\tau_i} r_i$ if $\tau_i < d_i$, and $w_i(\rho) = -M$ otherwise. The total reward collected on ρ is: $w(\rho) = \sum_{i=1}^m w_i(\rho)$. The *value* of the policy π, denoted V^π, is given as $V^\pi = \mathbb{E}_\rho^\pi[w(\rho)]$. Our baseline is the *optimal policy* π^* that maximizes the value, i.e., $\pi^* = \arg\max_\pi V^\pi$, and the *optimal value* is V^{π^*}.

Lower complexity bound: simplified setting with deterministic agent and stationary goals. We establish that the optimal policy synthesis problem is NP-hard, and we show this already on a simplified setting: Suppose the transition function T of the MO-MDP is deterministic, the goals are stationary such that the location of the i-th goal is always $g_i \in S$, and for any policy, the minimum travel time from a given location $x \in S$ to the i-th goal is $t_i = d(x, g_i)$. (Alternatively, t_i can be interpreted as the length of the shortest path from x to g_i when the agent moves at the unit velocity.)

Theorem 1. *Finding the optimal policy for the MO-MDP is NP-hard.*

Proof. The simpler setting with deterministic agents and stationary goals subsumes the Euclidean traveling salesman problem, where the deadlines are all infinity, and the goal is to find the shortest path passing through all the goals. This problem is known to be NP-complete [18], and our deadline-restricted case is at least as hard as it. **KM:** There is a slight difference, that we allow revisiting a state. However, I found on discussion forums that this does not change the complexity, find a reference. Also see metric traveling salesman, which is more general (we could then replace the Euclidean distance requirement in the MO-MDP state space with the requirement of just having a metric. □

Proposition 1. *Consider the simpler case of deterministic agent with stationary goals. There exists a permutation $\sigma^* \in \Sigma_m$, where Σ_m is the set of all permutations of $[1; m]$, such that the optimal policy π^* pursues the goals in the order $\sigma^*(1) \ldots \sigma^*(m)$, and π^* takes the shortest paths from the initial agent location to goal $\sigma^*(1)$ and between $\sigma^*(i)$ and $\sigma^*(i+1)$, for every $i \in [1; m-1]$.*

The above proposition effectively reduces the problem to an optimal *scheduling* problem. Furthermore, Theorem 1 suggests that a polynomial-time centralized optimal algorithm does not exist, unless P=NP, already for the simpler deterministic case with stationary goals. This motivates us to seek for a heuristic approach that is aimed towards finding the optimal schedule over the goals whose existence is guaranteed by Proposition 1.

A natural deterministic baseline is the *least slack time first* (LSTF) rule, which schedules the goal with smallest slack $slack_i(x, t) := d_i - t - d(x, g_i)$. However, this notion of urgency is tailored to stationary goals and deterministic travel times, and does not align with our full MO-MDP objective once targets move stochastically or may appear and disappear over time. For this reason, we use a more general benchmark that we call *reward-aware urgency first* (RAUF). **KM:** Is this used in the literature? If yes, some citation would be nice. At every decision point, RAUF selects one of the goals with maximal *reward-aware urgency*, namely the marginal continuation-value gain from granting that goal immediate control rather than deferring control to competing goals; ties are broken uniformly at random. Unlike slack, this urgency notion directly accounts for discounted rewards, missed-deadline penalties, stochastic target motion, and future target arrivals or removals. Our auction mechanism is designed to approximate this reward-aware urgency ordering in a decentralized manner while remaining tractable in settings where computing the globally optimal schedule is infeasible. **GS:** Also talk about how urgency encodes agent cooperation.

3 Auction-Based Multi-Policy Framework

We consider the compositional approach to policy synthesis for MO-MDPs, where we will design a selfish, *local* policy maximizing each individual value component, towards the fulfillment of some required global coordination requirements. The main crux is in the composition process, where each local policy may propose a different action, but the composition must decide one of the actions that will be actually executed. Importantly, the composition must be implementable in a distributed manner, meaning we will *not* use any global policy that would pick an action by analyzing all local policies and their reward functions.

We present a novel *auction*-based RL framework for compositional policy synthesis for MO-MDPs. In our framework, not only do the local policies emit actions, but they also *bid* for the privilege of executing their actions for a given number of time steps $\tau \in \mathbb{Z}_{>0}$ in future. The bids are all non-negative integers bounded by a given constant $\beta \in \mathbb{Z}_{\geq 0}$, and the highest bidder's actions get executed for the following τ consecutive steps, with bidding ties being resolved uniformly at random. To discourage policies from always bidding high, we penalize the policies proportional to their bids scaled with a constant $\rho \in (0, 1)$. We consider two intuitive alternatives for the penalty model, namely Winner-Pays and All-Pay: in the Winner-Pays setting, only the executed policy (one of the highest bidders) pays the penalty, whereas in the All-Pay setting, all policies pay penalties. Through this mechanism, policies select their bids to reflect how critical their action is in the current state, and the bidding penalty incentivizes truthful urgency. This way, we obtain a purely decentralized scheme to coordinate local policies in a given MO-MDP. **GS:** Maybe talk about existence of Nash equilibria here.

The parameters τ, β, and ρ denote the *bidding interval*, *maximum bid*, and *bid penalty coefficient*, respectively. Ablation studies in Section 5 highlight their importance. If τ is too small, policies switch too frequently, risking suboptimal outcomes. If β is too small, policies cannot express preference strength, leading to near-random control. If ρ is too large, policies tend to abstain from bidding.

Theoretical analysis of the bidding schemes. We now show that both Winner-Pays and All-Pay implement the reward-aware urgency ordering whenever one goal has a sufficiently large urgency advantage over the others. To capture stochastic motion of goals, we formulate the urgency scores using conditional expectations under the transition kernel of the MO-MDP. The deterministic stationary-goal case discussed above is then recovered as a special case in which these conditional expectations collapse to point values. **KM:** No need to talk about the deterministic case here.

Suppose π_i is the local policy for the i-th goal, mapping every state $\mathbf{s} = (s_0, s_1, \ldots, s_m) \in \mathbf{S}$ and time t to a probability distribution over the bid b_i and a deterministic action a_i. For convenience, we will write $\pi_i^b \colon (\mathbf{s}, t) \mapsto \alpha_i([0;\beta])$ and $\pi_i^a \colon (\mathbf{s}, t) \mapsto a_i$ as two separate components of the policy, where $\alpha_i([0;\beta])$ represents a probability distribution over $[0;\beta]$, and $b_i \sim \alpha_i([0;\beta])$ will represent a concrete bid value sampled from this distribution. Given $(\mathbf{s}, t)$, and given fixed π_i^a and π_i^b for all $i \in [1;m]$, we obtain a Markov chain, which is an MO-MDP with a singleton action set, with the deterministic initial state $\mathbf{s}$, i.e., $\mu(\mathbf{s}) = 1$, and each deadline d_i is replaced by $d_i - t$. This Markov chain induces a unique probability measure on the sample space of paths and rewards, and let $V_{\pi_i,\pi_{-i}}(\mathbf{s}, t)$ denote the *expected* discounted sum of reward obtained by the goal i in this Markov chain. (The notation π_{-i} indicates the collection of every j-th policy for $j \neq i$.) A policy profile $\{\pi_i\}_{i\in[1;m]}$ is called a *Nash equilibrium in the subgame starting at* $(\mathbf{s}, t)$ if for every i:

$$\forall \pi_i' \neq \pi_i \;.\; V_{\pi_i,\pi_{-i}}(\mathbf{s}, t) \geq V_{\pi_i',\pi_{-i}}(\mathbf{s}, t).$$

Suppose we are currently at $(\mathbf{s}, t)$, and let i be a given policy index. We consider two cases: (a) i wins the bidding, in which case i picks π_i^a to maximize its own reward, and the optimal reward for i is: $\widetilde{V}_i^{\mathsf{win}}(\mathbf{s}, t) \coloneqq \max_{\pi_i^a}$

Suppose π_i is the local policy for the i-th goal, mapping every agent state $s \in S$ and time t to a bid b_i and an action a_i. For convenience, we will write $\pi_i^b \colon (s, t) \mapsto b_i$ and $\pi_i^a \colon (s, t) \mapsto a_i$ as two separate components of the policy. At a bidding round (s, t), let $\Xi_i^{\mathsf{win}}(s, t)$ and $\Xi_i^{\mathsf{lose}}(s, t)$ denote the random future trajectories **KM:** for unbounded time in the future? induced by the environment and by the subsequent bids when objective i wins and loses the current auction, respectively. We first define the pre-payment continuation values

$$\widetilde{V}_i^{\mathsf{win}}(s, t) \coloneqq \mathbb{E}\big[w_i(\Xi_i^{\mathsf{win}}(s, t)) \mid s, t, i \text{ wins at } (s, t)\big]\,,$$
$$\widetilde{V}_i^{\mathsf{lose}}(s, t) \coloneqq \mathbb{E}\big[w_i(\Xi_i^{\mathsf{lose}}(s, t)) \mid s, t, i \text{ loses at } (s, t)\big]\,,$$

KM: Typo: s, t, i wins at (s, t) should be i wins at (s, t). **KM:** I do not understand the notation $w_i(\Xi_i^{\mathsf{win}}(s, t))$. where the expectation ranges over stochastic state transitions, target motion, target arrivals or removals, and future randomized bids and tie-breaking. **KM:** random arrival and removal of objectives is problematic, because this would cause the dimension of the state space to change, meaning the dimension of the underlying random variable (corresponding to "$\mathbb{E}$" to change. I would keep the number of goals fixed, this is a distraction and unnecessarily complicates the math. The post-payment values when objective i bids b_i are

$$
\begin{aligned}
V_i^{\mathsf{win}}(s,t,b_i) &:= \widetilde{V}_i^{\mathsf{win}}(s,t) - \rho b_i,\\
V_i^{\mathsf{lose}}(s,t,b_i) &:= \widetilde{V}_i^{\mathsf{lose}}(s,t) - \rho b_i \cdot \mathbf{1}[\mathsf{All\text{-}Pay}].
\end{aligned}
$$

The continuation value at a bidding state (s,t) is then

$$
\forall i \ .\ V_i^{\mathsf{cont}}(s,t) := \max_{b_i \geq 0} \mathbb{E}_{b_{-i}\sim\pi^b_{-i}(s,t)}[U_i(s,t,b_i,b_{-i})],
$$

KM: There is something wrong: $V_{\mathsf{win}}, V_{\mathsf{lose}}, \widetilde{V}_{\mathsf{win}}, \widetilde{V}_{\mathsf{lose}}$ do not depend on V_i^{cont}. where $b_{-i} \sim \pi^b_{-i}(s,t)$ denotes the collection of bids of all policies other than i **KM:** I think we need to say "...at the Nash equilibrium, ...", and

$$
U_i(s,t,b_i,b_{-i}) := \begin{cases} V_i^{\mathsf{win}}(s,t,b_i) & \text{if } b_i > \max_{j\neq i} b_j,\\ V_i^{\mathsf{lose}}(s,t,b_i) & \text{if } b_i < \max_{j\neq i} b_j,\\ \frac{1}{|W|} V_i^{\mathsf{win}}(s,t,b_i) + \left(1-\frac{1}{|W|}\right) V_i^{\mathsf{lose}}(s,t,b_i) & \text{if } i \in W, \end{cases}
$$

with $W := \{k \in [1;m] \mid b_k = \max_{\ell\in[1;m]} b_\ell\}$ being the set of highest bidders. The *net gain* for bidding b_i at (s,t) is defined as:

$$
\begin{aligned}
\Delta_i^{\mathrm{net}}(s,t,b_i) &:= V_i^{\mathsf{win}}(s,t,b_i) - V_i^{\mathsf{lose}}(s,t,b_i)\\
&= \Delta_i^{\mathrm{gross}}(s,t) - \rho b_i \cdot \mathbf{1}[\mathsf{Winner\text{-}Pays}],
\end{aligned}
$$

where the *gross gain* is

$$
\Delta_i^{\mathrm{gross}}(s,t) := \widetilde{V}_i^{\mathsf{win}}(s,t) - \widetilde{V}_i^{\mathsf{lose}}(s,t).
$$

The quantity $\Delta_i^{\mathrm{gross}}(s,t)$ is exactly the reward-aware urgency of policy i: it measures the continuation-value loss incurred by not allocating the next bidding interval to i. Thus RAUF selects a policy maximizing $\Delta_i^{\mathrm{gross}}(s,t)$.

The following theorem establishes that the $\mathsf{Winner\text{-}Pays}$ setting implements RAUF whenever the leading urgency exceeds the runner-up by more than one bid increment.

Theorem 2. *In the* $\mathsf{Winner\text{-}Pays}$ *setting, consider a single bidding round at the agent state $s \in S$ and time t. Suppose $\rho\beta > \max_{j\in[1;m]} \Delta_j^{\mathrm{gross}}(s,t)$,**KM:** I would make it $\rho\beta > r_{\max} + M$. It is always better to impose constraints on the hyper-parameters, because maintaining those constraints is in the designer's hand. However, a constraint on the $\Delta_j^{\mathrm{gross}}$ feels like a matter of "assumption" or "luck." and let i^* be a policy for which the following gross gain gap condition holds:*

$$
\Delta_{i^*}^{\mathrm{gross}}(s,t) - \max_{j\neq i^*} \Delta_j^{\mathrm{gross}}(s,t) > \rho. \tag{1}
$$

Then in any pure Nash equilibrium, policy i^* wins control.

Proof. Suppose for contradiction, some other policy $j \neq i^*$ wins control by bidding b_j, while policy i^* bids $b_{i^*} < b_j$. As this is a Nash equilibrium, it must hold that $\Delta_j^{\mathrm{net}}(s,t,b_j) > 0$, i.e., $\Delta_j^{\mathrm{gross}}(s,t) - \rho b_j > 0$, because otherwise, policy j would be better off by bidding zero to lose the bidding. This gives us the upper bound $\rho b_j < \Delta_j^{\mathrm{gross}}(s,t)$. It follows from $\rho\beta > \max_{k\in[1;m]} \Delta_k^{\mathrm{gross}}(s,t)$ that $\rho b_j < \rho\beta$, and hence $b_j < \beta$, i.e., the policy j does not make the highest possible bid.

Now we assumed that policy i^* had bid $b_{i^*} < b_j$, but we show that policy i^* would achieve a higher net gain by bidding $b_j + 1$, contradicting the Nash equilibrium: $\Delta_{i^*}^{\mathrm{net}}(s,t,b_j+1) = \Delta_{i^*}^{\mathrm{gross}}(s,t) - \rho(b_j+1) = \Delta_{i^*}^{\mathrm{gross}}(s,t) - \rho b_j - \rho > \Delta_{i^*}^{\mathrm{gross}}(s,t) - \Delta_j^{\mathrm{gross}}(s,t) - \rho > 0$. Here, notice that $b_j + 1$ is a valid bid since $b_j < \beta$, as we established above. $\square$

The previous theorem is abstract in the sense that it only assumes an urgency gap. The following lemma shows how the missed-deadline penalty $-M$ induces such a gap when delaying a target materially increases its probability of expiring.

Lemma 1 (Expiry-sensitive urgency gap)**.** *Fix a bidding state (s,t) and a policy i^*. Suppose there exists $\delta > 0$ such that* ***KM:*** *I think such a δ would always exist. We need to say "for every $\delta >$..."*

$$\Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ loses at } (s,t)] - \Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ wins at } (s,t)] \geq \delta,$$

KM: *i^* loses at (s,t)****KM:*** *I think τ_i^* is undefined. and suppose that for every $j \neq i^*$, granting or denying the current bidding interval to j does not change its probability of missing the deadline over the horizon****KM:*** *this I don't see, so that $\Delta_j^{\text{gross}}(s,t)$ is due only to changes in discounted reward timing. Then*

$$\Delta_{i^*}^{\text{gross}}(s,t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s,t) \geq \delta M - 2r_{\max}.$$

KM: One issue with this is that we promised earlier that unlike LSTF, our method is going to be "reward-aware." But this Lemma suggests that our method is "deadline-aware," isn't it?

Proof. By definition, the per-objective return can be written as

$$w_{i^*} = \gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] - M\,\mathbf{1}[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*}].$$

Substituting this identity into the definition

$$\Delta_{i^*}^{\text{gross}}(s,t) = \widetilde{V}_{i^*}^{\mathsf{win}}(s,t) - \widetilde{V}_{i^*}^{\mathsf{lose}}(s,t) = \mathbb{E}\left[w_{i^*}(\Xi_{i^*}^{\mathsf{win}}(s,t)) - w_{i^*}(\Xi_{i^*}^{\mathsf{lose}}(s,t)) \mid s,t\right],$$

and grouping the reward and penalty parts gives

$$\begin{aligned}\Delta_{i^*}^{\text{gross}}(s,t) = \mathbb{E}[\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] \mid s, t, i^* \text{ wins at } (s,t)] - \mathbb{E}[\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] \mid s, t, i^* \text{ loses at } (s,t)] \\ + M\Big(\Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ loses at } (s,t)] \\ - \Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ wins at } (s,t)]\Big).\end{aligned}$$

By assumption, the penalty term is at least δM. Moreover, since the truncated discounted reward term $\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}]$ always lies in $[0, r_{\max}]$, the difference between the two reward expectations for i^* is bounded below by $-r_{\max}$. Hence $\Delta_{i^*}^{\text{gross}}(s,t) \geq \delta M - r_{\max}$. For every $j \neq i^*$, the no-expiry-change assumption **KM:** make this assumption formal. implies that $\Delta_j^{\text{gross}}(s,t)$ contains only the discounted reward timing difference, and therefore $\Delta_j^{\text{gross}}(s,t) \leq r_{\max}$. Combining the two bounds yields

$$\Delta_{i^*}^{\text{gross}}(s,t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s,t) \geq \delta M - 2r_{\max}.$$

□

Corollary 1 (Winner-pays prioritizes near-expiry targets)**.** *In the* Winner-Pays *setting, suppose the assumptions of Lemma 1 hold at bidding state (s,t) for some policy i^*, and moreover*

$$\delta M - 2r_{\max} > \rho \qquad \textit{and} \qquad \rho\beta > \max_{j \in [1;m]} \Delta_j^{\text{gross}}(s,t).$$

Then in any pure Nash equilibrium, policy i^* wins control.

Proof. By Lemma 1,

$$\Delta_{i^*}^{\text{gross}}(s,t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s,t) \geq \delta M - 2r_{\max} > \rho.$$

Hence the gap condition (1) of the previous theorem holds, and the claim follows. □

The corollary should not be read as saying that M ought to be made arbitrarily large. Rather, it identifies a relative scale separation: stronger missed-deadline penalties enlarge the regime in which near-expiry targets are provably prioritized, provided the penalty signal dominates both the ordinary reward scale $r_{\max}$ and the bidding increment scale ρ. Choosing M therefore remains a modeling decision about how costly deadline misses should be in the underlying MO-MDP objective, not merely a theorem-tuning parameter.

For the All-Pay result below, we analyze the standard continuous-bid approximation of the auction game.**KM:** Why do you say this? This is justified when the bid grid is sufficiently fine and the maximum bid is sufficiently large, so that the discrete bounded game approaches the continuous all-pay contest.

The following theorem establishes that the All-Pay setting concentrates probability mass on the RAUF choice whenever the top urgency is strictly ordered above the remaining urgencies.

Theorem 3. *In the All-Pay setting with $m \geq 3$, consider a single bidding round at the agent state $s \in S$ and time t. Let i^* and j^* be policies for which the following gross gain order holds:*

$$\Delta_{i^*}^{\text{gross}} > \Delta_{j^*}^{\text{gross}} > \max_{k \notin \{i^*, j^*\}} \Delta_k^{\text{gross}}. \tag{2}$$

Then there is no Nash equilibrium with pure strategies, although a unique Nash equilibrium exists with mixed strategies (where at each (s,t), each policy uses a probability distribution to select the bid). In this Nash equilibrium, policy i^* wins control with probability $1 - \Delta_{j^*}^{\text{gross}}/2\Delta_{i^*}^{\text{gross}}$.

Proof. It follows from the work of Hillman and Riley [12] that in the Nash equilibrium, the optimal bid of the policies are as follows. The policy i^* picks a uniformly random bid b_i^* from $[0, \Delta_{j^*}^{\text{gross}}/\rho]$, the policy j^* picks a uniformly random bid b_j^* from $(0, \Delta_{j^*}^{\text{gross}}/\rho]$ while picking the bid 0 with probability $(\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}})/\Delta_{i^*}^{\text{gross}}$, and every other policy deterministically picks the bid 0. Clearly, if $b_i^* > b_j^*$, then i^* wins control, and we need to show that $P[b_i^* > b_j^*] = 1 - \Delta_{j^*}^{\text{gross}}/2\Delta_{i^*}^{\text{gross}}$. To show this, consider the probability density function $f_{i^*}(b) = \rho/\Delta_{j^*}^{\text{gross}}$ of b_{i^*}, and the cumulative distribution function $F_{j^*}(b) = (\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}})/\Delta_{i^*}^{\text{gross}} + \rho b/\Delta_{i^*}^{\text{gross}}$ of b_{j^*}. Then,

$$\begin{aligned}
P[b_i^* > b_j^*] &= \int_{-\infty}^{\infty} F_{j^*}(b) f_{i^*}(b) db \\
&= \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \left(\frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\rho b}{\Delta_{i^*}^{\text{gross}}} \right) \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db \\
&= \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db + \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \frac{\rho b}{\Delta_{i^*}^{\text{gross}}} \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db \\
&= \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot \frac{\Delta_{j^*}^{\text{gross}}}{\rho} + \frac{\rho^2}{\Delta_{i^*}^{\text{gross}} \Delta_{j^*}^{\text{gross}}} \cdot \frac{\left(\Delta_{j^*}^{\text{gross}}\right)^2}{2\rho^2} \\
&= \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}} \\
&= 1 - \frac{\Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}} \\
&= 1 - \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}}
\end{aligned}$$

□

If the second inequality in (2) becomes an equality, then it is known that the Nash equilibrium is no longer unique, and in fact, there will be a continuoum of Nash equilibria [5], and reaching the desirable equilibrium will require separate mechanisms. Instead, we argue that the event of any of the inequalities in (2) becoming an equality is a measure zero event, and therefore it is vanishingly unlikely in practice. Moreover, when $\Delta_{i^*}^{\text{gross}}$ is substantially larger than $\Delta_{j^*}^{\text{gross}}$, the winning probability

$$1 - \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}}$$

approaches 1. Hence, although All-Pay need not deterministically implement RAUF, it still concentrates control on the most reward-aware urgent policy whenever the top-two urgency gap is large.

Global stochastic approximation guarantee. The previous results are local one-step implementation statements. We now state an assumption-based guarantee showing that, under stochastic target motion and target appearance/disappearance, the auction policy can still achieve a constant fraction of the completions attained by an optimal scheduler over a finite horizon.

Let π^{auc} denote the policy induced by a Nash equilibrium of the bidding game, let $p^{\text{NE}}(s,t)$ be the distribution over winning objectives at bidding state (s,t), and let $N^{\text{auc}}(T)$ and $N^{\star}(T)$ be the numbers of completed goals by horizon T under π^{auc} and the optimal scheduler, respectively. Writing $u_i(s,t) \coloneqq \Delta_i^{\text{gross}}(s,t)$, **KM:** why define a new notation? define

$$\begin{aligned} u_{\max}(s,t) &\coloneqq \max_{i\in[1;m]} u_i(s,t), \\ H_\lambda(s,t) &\coloneqq \{i \in [1;m] \mid u_i(s,t) \geq \lambda u_{\max}(s,t)\}, \\ \beta(s,t) &\coloneqq \sum_{i \in H_\lambda(s,t)} p_i^{\text{NE}}(s,t), \end{aligned}$$

where $\lambda \in (0,1]$. **KM:** so p_i^{NE} is the i-th component of the distribution?

Assumption 1 (Urgency-completion bridge)**.** There exists $\kappa \in [0,1]$ such that for every reachable bidding state (s,t) and every objective i,

$$q_i(s,t) \geq \kappa \frac{u_i(s,t)}{u_{\max}(s,t)} q_{\max}(s,t), \qquad q_{\max}(s,t) \coloneqq \max_{i\in[1;m]} q_i(s,t),$$

where $q_i(s,t)$ is the conditional probability that objective i is eventually completed by horizon T **KM:** But this could result in missing the deadline d_i, right? if it is granted control at (s,t).

Assumption 2 (Time Regularity)**.** There exist constants $0 < \underline{\mu} \leq \overline{\mu} < \infty$ and $C_{\text{os}} \geq 0$ such that: (i) every attempt started under π^{auc} has conditional expected duration at most $\overline{\mu}$; (ii) no scheduler can achieve expected time per successful completion below $\underline{\mu}$; and (iii) the expected number of attempts started under π^{auc} before horizon T is at least $T/\overline{\mu} - C_{\text{os}}$.

Assumption 3 (Availability)**.** There exists $q_0 \in (0,1]$ such that at every reachable nonterminal bidding state (s,t),

$$q_{\max}(s,t) \geq q_0.$$

Theorem 4 (Stochastic fraction-of-optimal completion guarantee)**.** *Suppose the above three assumptions hold and $\beta(s,t) \geq \beta$ **KM:** Please don't use the same symbol to denote a function as well as a constant uniformly over reachable bidding states for some $\beta \in (0,1]$.**KM:** Is it the same β as the maximum bid? Then*

$$\mathbb{E}[N^{\text{auc}}(T)] \geq \lambda\beta\kappa q_0 \left(\frac{T}{\overline{\mu}} - C_{\text{os}}\right),$$

and moreover

$$\frac{\mathbb{E}[N^{\text{auc}}(T)]}{\mathbb{E}[N^\star(T)]} \geq \lambda\beta\kappa q_0 \left(\frac{\underline{\mu}}{\overline{\mu}} - \frac{C_{\text{os}}\underline{\mu}}{T}\right).$$

Proof. At any reachable bidding state (s,t), every objective in $H_\lambda(s,t)$ has urgency at least $\lambda u_{\max}(s,t)$. Hence

$$\sum_{i=1}^m p_i^{\text{NE}}(s,t) u_i(s,t) \geq \lambda\beta(s,t) u_{\max}(s,t) \geq \lambda\beta u_{\max}(s,t).$$

By the urgency-completion bridge, the conditional probability that the auction-selected objective is eventually completed satisfies

$$\sum_{i=1}^m p_i^{\text{NE}}(s,t) q_i(s,t) \geq \kappa \frac{q_{\max}(s,t)}{u_{\max}(s,t)} \sum_{i=1}^m p_i^{\text{NE}}(s,t) u_i(s,t) \geq \lambda\beta\kappa q_{\max}(s,t) \geq \lambda\beta\kappa q_0.$$

Summing this lower bound over the expected number of attempts started before horizon T and using the time-regularity assumption yields

$$\mathbb{E}[N^{\text{auc}}(T)] \geq \lambda\beta\kappa q_0 \left(\frac{T}{\overline{\mu}} - C_{\text{os}}\right).$$

On the other hand, the lower bound $\underline{\mu}$ on expected time per successful completion implies $\mathbb{E}[N^\star(T)] \leq T/\underline{\mu}$. Combining the two inequalities gives the claimed ratio bound. □

Remark 1 (Why λ can be chosen large in practice). The factor λ measures how close the selected "high-urgency" objectives are to the maximum reward-aware urgency. In deadline-driven environments, a natural choice is to define the urgent set through near-expiry targets. Suppose each target remains active for a lifetime L after appearing, let μ denote a typical completion-time scale, and call a target deadline-urgent when its remaining lifetime is at most $c\mu$ for some constant $c > 0$. When target ages are approximately asynchronous and spread across $[0, L]$, the fraction of such near-expiry targets is on the order of $\min\{1, c\mu/L\}$. For these targets, delaying service by even one bidding interval causes a substantial increase in expiry probability, so the missed-deadline term contributes on the order of M to their urgencies. By contrast, non-urgent targets have urgency variation dominated by the discounted reward scale $r_{\max}$. Consequently, in regimes where $M \gg r_{\max}$ and expiries are the main source of urgency, the objectives that matter for the approximation bound are not merely above average; they are typically close to the maximizer of $u_i(s, t)$, which makes choosing λ near 1 plausible.

Remark 2. Theorem 4 is stated for completion counts because this is the most robust quantity under stochastic target motion and dynamic target sets. When per-goal rewards are uniformly bounded away from zero and missed-goal penalties admit a common additive budget, the same theorem immediately yields a corresponding constant-factor guarantee for the total reward objective up to the associated reward-scaling and additive-shift terms.

4 Learning Local Policies via Markov Games

We formalize our framework as an m-player general-sum Markov game [16], where each player optimizes one of the m objectives of the original MO-MDP. The game encodes the bidding mechanism in its transitions and rewards, and we therefore call it a *Markov bidding game*. Each state records the current MO-MDP state, the policy selected in the last bidding round, and the remaining time until the next bidding. At each state, players independently choose an action and a bid; transitions determine the winning policy based on the bids, while rewards capture the corresponding bidding penalties.

Definition 1 (Markov bidding games)**.** Let $\mathcal{M} = (S, A, T, R, \gamma, \mu)$ be an MO-MDP with m objectives, τ be the bidding interval, β be the maximum bid, ρ be the bid penalty coefficient, and $\Delta \in \{\textsf{Winner-Pays}, \textsf{All-Pay}\}$ be the used penalty model. We define the m-player Markov bidding game as a tuple $\mathcal{G}^{\mathcal{M}}_{\tau,\beta,\rho} = (\hat{S}, \hat{A}, \hat{T}, \hat{R}, \gamma, \hat{\mu})$ where

- $\hat{S} = S \times [1; m] \times [0; \tau - 1]$ is the state space. An element of $\hat{S}$ is denoted (s, k, t).
- $\hat{A} = (A \times \mathbb{Z}_{\geq 0})^m$ represents the joint extended action space of the m players, where each player independently selects an action from A and a nonnegative bid from $\mathbb{Z}_{\geq 0}$. We denote a set of actions chosen by the m players with $(\mathbf{a}, \mathbf{b}) = (a_i, b_i)_{i \in [1;m]} \in \hat{A}$.
- $\hat{T} : \hat{S} \times \hat{A} \times \hat{S} \to \mathbb{R}$ is the new transition function defined as,

$$\hat{T}((s,k,t), (\mathbf{a}, \mathbf{b}), (s', k', t')) \coloneqq \begin{cases} T(s, a_{k'}, s') & t > 0 \wedge t' = t - 1 \wedge k' = k, \\ \frac{1}{|\mathbf{b}_{\max}|} T(s, a_{k'}, s') & t = 0 \wedge t' = \tau - 1 \wedge k' \in \mathbf{b}_{\max}, \\ 0 & \text{otherwise}, \end{cases}$$

  where $\mathbf{b}_{\max} \coloneqq \{i \mid b_i = \max\{b_1, \ldots, b_m\}\}$ is the set of agents with the highest bid.
- $\{\hat{R}_i : \hat{S} \times \hat{A} \times \hat{S} \to \mathbb{R}\}_{i \in [1;m]}$ is the set of reward functions for the m agents with

$$\hat{R}_i((s,k,t), (\mathbf{a}, \mathbf{b}), (s', k', t')) \coloneqq \begin{cases} R_i(s, a_{k'}, s') - \rho \cdot b_i & t = 0 \wedge (i = k' \vee \Delta = \textsf{All-Pay}), \\ R_i(s, a_{k'}, s') & \text{otherwise}. \end{cases}$$

- $\hat{\mu}$ is the initial state distribution over $\hat{S}$ defined as,

$$\hat{\mu}(s, k, t) \coloneqq \begin{cases} \mu(s) & t = 0 \wedge k = i, \\ 0 & \text{otherwise}, \end{cases}$$

  where $i \in [1; m]$ is an arbitrary agent.

In the Markov game $\mathcal{G}^{\mathcal{M}}_{\tau,\beta,\rho}$, each player i aims to compute its own selfish local policy of the form $\pi_i\colon (\hat{S} \times \hat{A})^* \times \hat{S} \mapsto (a_i, b_i) \in A \times \mathbb{Z}_{\geq 0}$, such that the discounted sum of its own reward is maximized, given that the other players are selfishly fulfilling their own objectives.

Learning policies in Markov bidding games. Since the transitions of the original MO-MDP are unknown, the transitions of the constructed Markov game in Definition 1 will also be unknown. Our goal is to learn the local policy of each player using RL. **(author?)** [27] demonstrate that proximal policy optimization (PPO; [23]) can be effective to train multiple policies concurrently for multi-agent settings. We follow their recipe and train all the local policies with PPO. Similar to **(author?)** [27], since the objectives in our environments share the same characteristics, we use the same actor and critic network for all the policies. At each iteration of PPO, we collect all the rollouts from each policy's perspective and sample minibatches from this pool to perform the PPO updates.

Additionally, to cope with varying number of objectives and varying number of players in the resulting Markov game, we employ the *attention pooling* architecture to compress any number of objectives into a fixed-sized embedding. This design enables the agent to operate with an arbitrary number of policies (i.e., objectives) at deployment time. Each objective information vector is first processed by a fully connected neural network to produce a target embedding. These embeddings are then aggregated through a weighted sum, where the weights are computed by taking the dot product between each embedding and a learned query vector. This attention mechanism produces a fixed-dimensional representation that summarizes the set of objectives while allowing the model to adapt to varying objective counts.

5 Experimental Results

We conduct experiments to show the superior performance of our method compared to the baselines. We also perform ablations with respect to all the key components of the framework; and draw insights on their importance and how we can tune them. A detailed description of the environments and the baseline approaches is provided in Appendix A, below we include a summary.

Figure 2: Atari Assault env.

Environments. We evaluate our approach on two environments: a mobile cat feeder, introduced in Section 1, and the Atari game Assault. In the Cat Feeder environment, a robot moves on a 30×30 grid to feed up to m cats that move according to unknown stochastic processes. Each cat disappears after a fixed number of steps; feeding it yields a reward, while expiration incurs a penalty of equal magnitude. We train with $m = 7$ cats and test with up to 14. Each cat is served by a dedicated policy whose observation includes the robot position, the locations of active cats, and their remaining lifetimes.

In Assault (Figure 2), the agent controls a missile launcher to destroy incoming alien ships while avoiding enemy missiles [6]. We treat each ship as a separate objective and assign a local policy to pursue it, enabling a modular and interpretable design. Destroying a target yields reward, losing a life incurs a penalty, and performance is measured using the game score.

We also consider two variants of each environment: one without the attention pooling module (only for Cat Feeder; we do not use attention pooling for Atari Assault since there are only up to 3 targets) and another in which the local policies can only observe their own objectives (denoted Local Obs). Additionally, the supplementary material includes our codebase and recorded rollouts of our bidding-based policies on the two environments.

Baseline approaches. We compare against two baselines: a single PPO policy with aggregated rewards and deep W-learning (DWN) [21]. The PPO baseline is implemented using CleanRL [13]; for the Cat Feeder environment (with variable objective count) we additionally test sparse rewards and two shaping strategies. DWN trains one DQN policy per objective together with W-networks that estimate urgency scores of states. Hyperparameters are provided in Appendix B, with Cat Feeder tuned using Optuna [1].

Learning dynamics. Figure 3 shows the learning curves for multi-policy bidding and single-policy PPO algorithms on the two environments. While the curves are more noisy with Atari Assault,

Table 1: Performance (mean and standard deviation across 5 seeds) at the best evaluation iteration. Cat Feeder values measure performance as the difference between number of feeding requests completed and those that expired. Atari Assault values are the game score.

Method	**Cat Feeder**	**Atari Assault**
Winner-Pays	51.17 (4.49)	**663.92** (23.73)
w. Local Obs	67.21 (5.75)	644.00 (28.31)
w. No Attn. Pool.	58.41 (5.09)	—
All-Pay	**79.46** (3.68)	**668.48** (21.34)
w. Local Obs	69.73 (3.57)	648.72 (53.85)
w. No Attn. Pool.	48.78 (47.10)	—
DWN	−2.28 (2.22)	310.80 (159.54)
Single-Policy PPO		**430.08** (16.43)
w. No Reward Shaping	−48.75 (4.30)	—
w. Nearest Shaping	**22.17** (36.34)	—
w. Expiry Shaping	−44.34 (1.38)	—

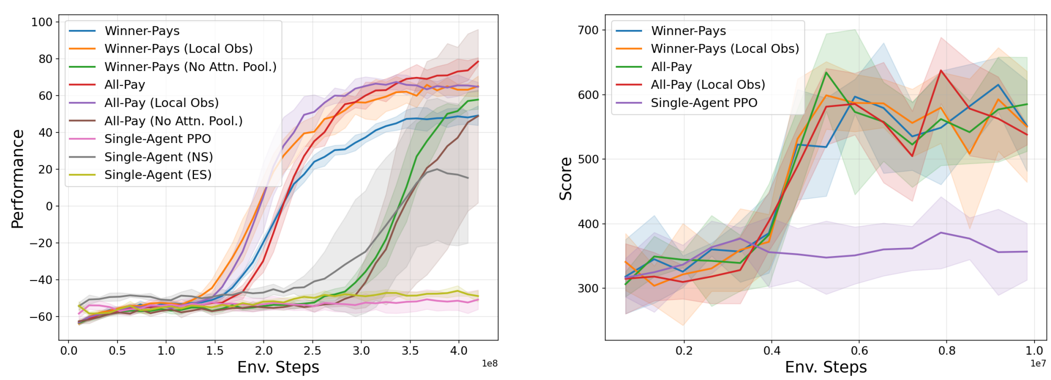
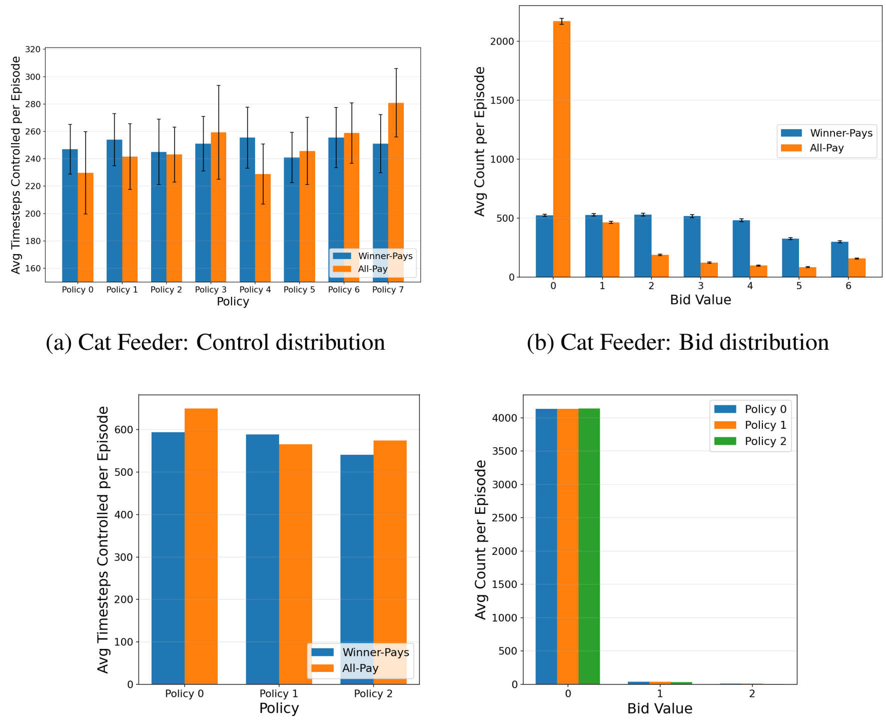


Figure 3: Learning curves for different methods on Cat Feeder (left) and Atari Assault (right). We plot mean over 5 seeds at each evaluation iteration and the shaded region is $\pm$ standard deviation.

they exhibit clearly distinguishable sigmoidal shapes with Cat Feeder. The curves for All-Pay and Winner-Pays methods start growing around the same time. Meanwhile, the variants with only local observations for each policy start growing much earlier. The variants with no attention pooling for targets only start growing much later. This validates the use of the attention pooling architecture to learn useful embeddings of targets. We remark that, based on the multi-agent PPO recipe of **(author?)** [27], we also found that using a small clip coefficient ($\sim$0.05) and large batch sizes was crucial to obtain slower but stable convergence of the policies towards a joint-equilibrium.

Comparison with baselines. Table 1 lists the performance of our method as well as the baselines on both environments. Our multi-policy methods outperform the baselines by a substantial margin. Across both the environments, we speculate that the poor performance of the single-policy PPO learner is due to the difficulty of credit assignment over a rollout in which there are multiple objectives that provide a positive reward. Additional reward shaping only provides moderate gains. In the Atari Assault environment, there is a delay between shooting a missile and it reaching its target. Since we only supply information from one frame in the observation, the learner could find it hard to associate the action of shooting and the reward from the death of a ship, especially if there is subsequent firing. On the other hand, the credit assignment becomes easier when we separate the objectives and each learner only has to focus on one target.

DWN performs poorly in all experiments. We recall that this method learns a W-network trained to predict the negative TD-error for each policy when the policy does not play its action. However, in our environments, we have a long horizon and all the objectives are symmetric with similar Q-values. This means that the TD-error is small and does not provide a strong signal to choose between policies at most timesteps. In contrast, our approach allows the policies themselves to indicate the importance of playing an action.



(a) Cat Feeder: Control distribution

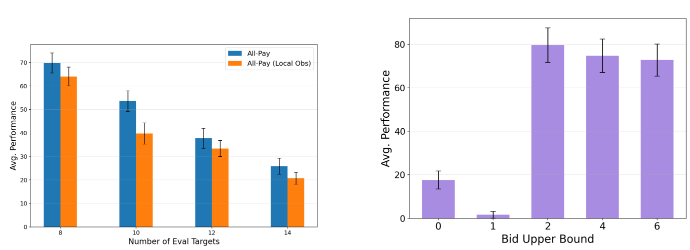
(b) Cat Feeder: Bid distribution

(c) Atari Assault: Control distribution

(d) Atari Assault: Bid Distribution (Winner-Pays)

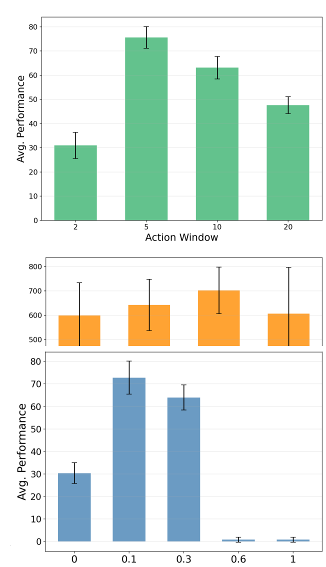
Figure 4: Comparison of control and bid distributions across environments and methods. The control distribution is the number of timesteps controlled by each policy, while the bid distribution is the distribution over bids picked by the policies.

Interpretability of policies and bidding behaviors. Video rollouts show that bidding-based policies take control when they can accumulate reward within the action window, while each policy pursues only its own objective. Figure 4 presents the distribution of control time and bids. Control is well distributed among policies. In Assault, bids are usually 0, indicating that random control assignment suffices since targets do not expire. In contrast, the Cat Feeder environment shows a wider bid spread (0–6). Here, All-Pay yields more conservative policies with mostly low bids, while Winner-Pays produces a more even distribution, reflecting greater willingness to bid high.



(a) Scaling number of targets.

(b) Ablation: Bid upper bound



(c) Ablation: Bid penalty

Figure 5: Scaling number of targets and ablations on the Cat Feeder environment.

Scaling number of targets. We conduct an experiment with the Cat Feeder environment in which the number of cats seen during testing is larger than that during training. Figure 5a shows that the policies continue to perform well up to 14 targets even though they were only trained to compete with up to 7 other policies. The attention pooling mechanism plays a critical role as it can generate useful embeddings of an arbitrary number of competing targets.

Further ablations. Figure 5b shows that at least two bid levels are needed for good performance; with only one level, policies cannot compete effectively. Figure 5c evaluates the bid penalty factor: without penalties all policies overbid, hurting performance, while overly large penalties discourage bidding. Finally, Figure 6 studies the action window size, revealing an optimal value. Smaller windows cause excessive switching between policies, while larger ones prevent sufficient switching to serve all objectives.

6 Conclusion and Future Directions

We consider the problem of adapting RL policies due to evolving objectives, where any number of objectives may appear or disappear at runtime, and the agent must adapt as swiftly as possible. We present a compositional solution where each objective is served using a dedicated local policy, and each local policy competes against the other policies through an auction-based mechanism to select its preferred action. The training is formulated as a nonzerosum game between the local policies, and it is implemented using PPO in our framework. Using two challenging benchmarks, we demonstrate that our framework produces interpretable policies that achieve superior learning rate and dramatic performance improvement compared to the baseline approaches.

While we have only considered multi-objective environments in which the objectives belong to the same family, we will extend the framework to combinations of objective classes. The difficulty lies in the fact that a policy learned to compete against opponents from a given class may not work against opponents for a novel class of objectives seen at runtime. Additionally, richer bidding mechanisms might be necessary. For instance, while combining reachability and safety objectives, usually safety needs to get the priority when there is a tie in the bidding, instead of a random resolution.

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A Experimental Setup (Detailed)

We consider two multi-objective environments for our empirical studies, namely a mobile cat feeder and the Atari game called Assault. These environments are described below.

Environment I: Cat Feeder. The environment is as described in Section 1 and is modeled using a 30×30 grid with up to m cats that are moving continuously according to unknown stochastic processes. The value of m was set as 8 during the training time, whereas it was varied up to 14 during the testing time. Each cat disappears after a given number of time steps. If the agent reaches a cat before it disappears, it receives a reward, and otherwise it receives a penalty of equal magnitude. To avoid sparsity of the reward function, we provide each policy a small reward for going closer to its target. Performance is measured as the difference between the number of feeding requests completed and the number of those that expired.

In our modular framework, each cat will be chased by a dedicated local policy. The observation space of each local policy includes the current position of the robot, the current locations of the active requests, and the time steps until the expiration of the requests.

Environment II: Atari Assault. We consider the Atari game called Assault, where the agent controls a missile launcher to shoot oncoming alien ships while avoiding enemy missiles. The standard approach is to view this as a single-objective environment, where the agent's policy observes the entire image or the 128 byte raw RAM state observations to determine the optimal actions [6].

For our modular framework, we break the objective down to a multi-objective problem, where each alien ship is viewed as a separate objective, and each local policy attempts to destroy its assigned target while avoiding enemy missiles. Not only does this facilitate an improvement in the performance, as we will demonstrate, but also we obtain a transparent design, because which target is being chased is revealed at each point. This modular design is made possible by the object-centric variant of the Atari environment [8], which exposes the states of each individual target and the missiles by decoding the RAM states. The full observation space of each local policy also include the current position of the missile launcher, the current health, and the number of remaining lives. The local policy gets a reward or a penalty upon destroying its target enemy ship or losing a life, respectively. Lastly, we use the game score as the metric for evaluation.

Baseline approaches. We compare our method against two baselines: a single policy PPO with weighted rewards and deep W-learning (DWN; **(author?)** [21]). For the PPO baseline, since all the objectives have the same importance across the two environments, we use the total reward across objectives at each step as the reward function. We note that in the Atari Assault environment we do not use the game score as a reward and instead use the same rewards perceived by the bidding-based policies to ensure a fair comparison. The observation space remains the same as the multi-policy setting. In the Cat Feeder environment, we implement three reward mechanisms to help guide learning: one only with the sparse rewards from the environment, one with distance-based dense reward with respect to the nearest feeding request (denoted Nearest Shaping/NS in the experiments), and finally one where we provide distance-based reward with respect to the request that has the least amount of time until expiration (denoted Expiry Shaping/ES). The PPO implementation is taken from CleanRL [13]. We use the specified PPO hyperparameters from CleanRL for Atari Assault and we tune the hyperparameters for the Cat Feeder environment with 40 Optuna iterations [1].

DWN is another multi-policy method for multi-objective RL that uses DQN for each of the local policies and additionally learns a W-network for each policy that predict scalar weights. These weights are used to choose which policy's action should be played at each step. The W-network is trained alongside the Q-network by updating it with a gradient step to predict the negative temporal difference (TD) error when the policy does not get to play its action. The assertion here is that the negative TD-error quantifies how important it is for the policy to play its action. The implementation is borrowed from the repository of **(author?)** [21]. All hyperparameters for the baselines are listed in Appendix B.

B Reproducibility Details

The code for all experiments is included in the supplementary materials.

All experiments use 5 random seeds: {1825, 410, 4507, 4013, 3658}.

B.1 Cat Feeder Environment

Parameter	Value
Grid size	30×30
Number of agents / targets	8
Target reward	50.0
Target expiry steps	200
Target expiry penalty	50.0
Max episode steps	2000
Moving targets	True
Direction change probability	0.1
Target move interval	5

Table 2: Cat Feeder shared (training) environment parameters.

Parameter	Value
Auction	
Bid upper bound	6
Bid penalty	0.1
Action window	5
Distance reward scale	0.6
Training	
Iterations	400
Parallel environments	4096
Steps per rollout	256
Minibatches	256
PPO epochs	4
Learning rate	2.5×10^{-4} (annealed)
Discount γ	0.99
GAE λ	0.95
Clip coefficient	0.05
Entropy coefficient	0.03
Value function coefficient	1.0
Max gradient norm	0.5
Network	
Actor hidden layers	$[128, 128, 128, 128]$
Critic hidden layers	$[256, 256, 256, 256]$
Target attention pooling	True
Target embedding dim	64
Target encoder hidden layers	$[64, 64]$

Table 3: Cat Feeder multi-agent PPO hyperparameters (All-Pay and Winner-Pays).

Parameter	Value
Training	
Iterations	400
Parallel environments	4096
Steps per rollout	256
Minibatches	512
PPO epochs	8
Learning rate	1.74×10^{-4} (annealed)
Discount γ	0.963
GAE λ	0.970
Clip coefficient	0.327
Entropy coefficient	1.03×10^{-4}
Value function coefficient	1.076
Max gradient norm	0.840
Distance reward scale	0.0
Target attention pooling	False
Network (shared with multi-agent PPO)	
Actor hidden layers	$[128, 128, 128, 128]$
Critic hidden layers	$[256, 256, 256, 256]$

Table 4: Cat Feeder single-agent PPO baseline hyperparameters. Hyperparameters were tuned using Optuna with 40 trials. The nearest-target shaping variant is identical but uses a distance reward scale of 0.6 with shaping toward the nearest target.

Parameter	Value
Training	
Total timesteps	5×10^7
Parallel environments	256
Discount γ	0.99
Replay buffer size	1×10^6
Batch size	256
Learning starts	1×10^5
Train frequency	2560
W-network train delay	1×10^6
Target network update frequency	1000
Target network τ	1.0
Networks	
Q-network hidden layers	$[256, 256, 256, 256]$
W-network hidden layers	$[128, 128, 128]$
Q-network learning rate	1×10^{-4}
W-network learning rate	1×10^{-4}
Epsilon schedules	
Q ε start / min / decay	0.99 / 0.01 / 0.99
W ε start / min / decay	0.99 / 0.01 / 0.99

Table 5: Cat Feeder DWN baseline hyperparameters.

651 ### B.2 Assault Environment

Parameter	Value
Number of agents	3
Max enemies	3
Max episode steps	10000
Enemy destroy reward	10.0
Life loss penalty	10.0
Fire-while-hot penalty	8.0
Action window	15
Bid upper bound	2
Bid penalty	0.3

Table 6: Assault shared environment parameters.

Parameter	Value
Training	
Iterations	150
Parallel environments	128
Steps per rollout	512
Minibatches	8
PPO epochs	8
Learning rate	1×10^{-4} (annealed)
Discount γ	0.99
GAE λ	0.95
Clip coefficient	0.05
Entropy coefficient	0.05
Value function coefficient	0.5
Max gradient norm	0.5
Target KL	0.02
Network	
Actor hidden layers	$[128, 128, 128, 128]$
Critic hidden layers	$[256, 256, 256, 256]$

Table 7: Assault multi-agent PPO hyperparameters (All-Pay and Winner-Pays).

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Parameter	Value
Learning rate	2.5×10^{-4} (annealed)
Clip coefficient	0.1
Entropy coefficient	0.01
Target KL	—

Table 8: Assault single-agent PPO baseline hyperparameters (differences from multi-agent PPO only; all other parameters are shared).

Parameter	Value
Training	
Total timesteps	1×10^7
Parallel environments	8
Discount γ	0.99
Replay buffer size	5×10^5
Batch size	256
Learning starts	1×10^4
Train frequency	80
W-network train delay	1×10^5
Target network update frequency	1000
Target network τ	1.0
Networks	
Q-network hidden layers	$[256, 256, 256]$
W-network hidden layers	$[128, 128, 128]$
Q-network learning rate	1×10^{-4}
W-network learning rate	1×10^{-4}
Epsilon schedules	
Q ε start / min / decay	0.99 / 0.01 / 0.995
W ε start / min / decay	0.99 / 0.01 / 0.995

Table 9: Assault DWN baseline hyperparameters.

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